



Full length article

Positive allosteric modulation of GABA_A receptors by a novel antiepileptic drug cenobamate

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ABSTRACT

Cenobamate is a novel antiepileptic drug under investigation for use in patients with focal (partial-onset) seizures. To understand its potential molecular mechanism of action, the effects of cenobamate on GABA_A-mediated currents and GABA_A receptors in rodent hippocampal neurons were examined. Cenobamate potentiated GABA-induced currents (I_{GABA}) in acutely isolated CA3 pyramidal cells in a concentration-dependent manner (EC_{50} , 164 μM), which was not affected by flumazenil, a benzodiazepine receptor antagonist. Cenobamate enhanced tonic GABA_A currents (I_{tonic}), which is defined as a holding current shift by the GABA_A receptor antagonist bicuculline (EC_{50} , 36.63 μM). At therapeutically relevant concentrations, cenobamate induced minimal changes in the frequency, amplitudes, and decay time of spontaneous inhibitory postsynaptic currents in the CA1 neurons. Flumazenil failed to affect cenobamate-potentiated I_{tonic} and I_{phasic} in CA1 neurons. Cenobamate showed positive allosteric modulation of GABA-induced I_{GABA} mediated by GABA_A receptors. This effect was similar for all tested hGABA_A receptors containing six different alpha subunits ($\alpha_1\beta_2\gamma_2$ or $\alpha_{2.6}\beta_3\gamma_2$), with EC_{50} values ranging from 42 to 194 μM . Cenobamate did not displace the binding of flunitrazepam, a benzodiazepine derivative, or flumazenil to GABA_A receptors. The results showed that cenobamate, a novel antiepileptic drug, acts as a positive allosteric modulator of high-affinity GABA_A receptors, activated by GABA at a site independent of the benzodiazepine binding site and efficiently enhances I_{tonic} inhibition in hippocampal neurons, which could be an underlying molecular mechanism stabilizing neural circuits of the epileptic hippocampus.

1. Introduction

γ -Aminobutyric acid (GABA) is the primary inhibitory neurotransmitter in the mammalian brain, and its principal role is to reduce neuronal excitability by maintaining an inhibitory tone. Failure to maintain the balance between excitatory and inhibitory transmission within neural circuits is thought to be associated with epilepsy. There is growing evidence that GABA plays an important role in epileptogenic and antiepileptic mechanisms (Treiman, 2001). Animal models of genetic and acquired epilepsy provide evidence of an underlying impairment in GABAergic function in the brain (Bessaih et al., 2006; Reddy and Kuruba, 2013; Treiman, 2001; van Luijckelaar and Sitnikova, 2006). GABA levels were shown to be reduced in brain tissue and in

cerebrospinal fluid of patients with epilepsy (Meldrum, 1989; Petroff et al., 1996), and pharmacologic approaches to increasing GABAergic inhibition, such as the use of GABA agonists (muscimol) or positive allosteric modulators (benzodiazepines and barbiturates), have demonstrated anticonvulsant action, while GABA antagonists (bicuculline) produce seizures (Meldrum, 1989; Pfeiffer et al., 1996; Treiman, 2001). Thus, the enhancement of GABA-mediated neuronal inhibition is an effective mechanism of action of antiepileptic drugs.

Cenobamate ([*(R)*-1-(2-chlorophenyl)-2-(2H-tetrazol-2-yl)ethyl] carbamate) is a novel antiepileptic drug under investigation for use in patients with focal (partial-onset) seizures. Preclinical seizure models in rodents have shown that cenobamate exhibits a broad spectrum of anticonvulsant activity (Bialer et al., 2013). In clinical studies of

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patients with uncontrolled focal seizures, adjunctive therapy with cenobamate significantly reduced seizure frequency compared with placebo (Chung et al., 2014) and was associated with high rates of seizure freedom (Krauss et al., 2020). Impaired neuronal inhibition due to GABAergic dysfunction in the brain has been demonstrated as a critical epileptic mechanism (Cossart et al., 2001; Kobayashi and Buckmaster, 2003; Leroy et al., 2004). It is possible that the anti-epileptic action of cenobamate is, at least partially, mediated through positively regulating GABAergic inhibition in the brain, although these effects have not been thoroughly investigated.

GABA released from synaptic vesicles activates GABA_A receptors located at synaptic contacts, generating a transient or phasic current classically named inhibitory postsynaptic currents (IPSCs) or I_{phasic} . Similarly, low concentrations of ambient GABA can persistently activate certain subtypes of high-affinity GABA_A receptors, which are often remote from synapses (peri- or extra-synaptic receptors), generating a tonic conductance (I_{tonic}) (Farrant and Nusser, 2005; Semyanov et al., 2004). The dual mode of GABA_A-mediated inhibition has been originally reported and extensively studied in the dentate gyrus and principal neurons of the hippocampus (Pandit et al., 2013; Prenosil et al., 2006). The hippocampus is the brain region involved with memory, and it is implicated in the generation and perpetuation of certain types of seizures. The present study examines the effects of cenobamate on GABAergic neurotransmission, specifically its effects on GABA_A receptors mediating I_{phasic} and I_{tonic} in the principal neurons of the hippocampus.

2. Materials and methods

2.1. Animals

For the electrophysiology studies, rats were purchased from Samtako Bio (Kyung Gi-Do, Korea) and were housed under a 12/12-h (h) light/dark schedule. Animals were allowed free access to food and water until used. All experiments were approved by the Chungnam National University IACUC (Registration Number: CNU-00847) and complied with the guiding principles for the care and use of animals approved by the Council of the Physiological Society of Korea and/or the National Institutes of Health Guide for the Care and Use of Laboratory Animals.

2.2. Electrophysiology of isolated single CA3 neurons

For CA3 single-cell isolation, Sprague Dawley rats (12–17 days old, either sex) were decapitated under ketamine anaesthesia (100 mg/kg, i.p.). The brain was dissected and sliced transversely at a thickness of 400 μm using a microslicer (VT1000S; Leica, Nussloch, Germany). Slices containing the hippocampus were kept in an incubation medium (in mM: 124 NaCl, 3 KCl, 1.5 KH_2PO_4 , 24 NaHCO_3 , 2 CaCl_2 , 1.3 MgSO_4 , and 10 glucose; pH about 7.45) saturated with 95% O_2 and 5% CO_2 at room temperature (22–24 °C) for at least 1 h before the mechanical dissociation. Details of the mechanical dissociation have been described previously (Akaike and Moorhouse, 2003; Rhee et al., 1999). Briefly, mechanical dissociation was accomplished using a custom-built vibration device and a fire-polished glass pipette oscillating at about 50–60 Hz (0.3–0.5 mm) on the surface of the CA3 pyramidal layer. Slices were removed and mechanically dissociated neurons were allowed to settle and adhere to the bottom of the dish for 15 min. These dissociated neurons lost their most distal processes, but a short portion (~100 μm in a length) of their thick proximal dendrites was retained (Jang et al., 2006; Nakamura et al., 2019).

All electrophysiological measurements were recorded using conventional whole-cell patch-clamp techniques and a standard patch-clamp amplifier (Axopatch 200B; Molecular Devices, Union City, CA, USA). Neurons were clamped at a holding potential of 0 mV. The resistance of the recording pipettes filled with internal solution (in mM: 140 Cs-methanesulfonate, 10 CsCl, 2 EGTA, 4 ATP-Mg, and 10 HEPES

with the pH adjusted to 7.2 with Tris-base) was 1.5–2.5 M Ω . Membrane currents were filtered at 2 kHz (Axopatch 200B), digitized at 10 kHz, and stored on a computer equipped with pCLAMP 10 software (Molecular Devices). All experiments were performed at room temperature (22–25 °C).

The amplitudes of GABA-induced currents (I_{GABA}) were calculated by subtracting the baseline from the peak amplitude. The effects of cenobamate or diazepam were quantified as percentage changes compared to the control values.

2.3. Electrophysiology of hippocampal slices

Patch-clamp recordings were obtained in CA1 neurons of acutely prepared coronal hippocampal slices as previously described (Pandit et al., 2013). Briefly, brains were rapidly extracted from 18- to 21-day-old male Wistar rats anesthetized with tribromoethanol (Avertin, 200 mg/kg, i.p.). Slices were perfused with artificial cerebrospinal fluid (in mM: NaCl 126, KCl 2.5, MgSO_4 1, NaHCO_3 26, NaH_2PO_4 1.25, glucose 20, ascorbic acid 0.4, CaCl_2 1, pyruvic acid 2; pH 7.3–7.4; saturated with 95% O_2 –5% CO_2). Recordings were obtained using an Axopatch 200B amplifier. Current output was filtered at 2 kHz and digitized at 10 kHz (Digidata 1322A). Patch pipettes were filled with a high Cl^- -containing solution (in mM: KCl 140, HEPES 10, Mg^{2+} ATP 5, MgCl_2 0.9, and EGTA 10).

Spontaneous inhibitory postsynaptic currents (sIPSCs, recorded at –70 mV) were detected and analyzed using Mini Analysis (Synaptosoft, Decatur, GA). The decay phase of IPSCs best fitted with a double-exponential function was presented in weighted values and compared before and during the application of cenobamate (Pandit et al., 2014). The currents were recorded in the presence of 6-cyano-7-nitroquinoxaline-2, 3-dione (10 μM) and 2-amino-5-phosphonopivalic acid (APV, 100 μM) to isolate IPSCs. A stock solution of 6-cyano-7-nitroquinoxaline-2, 3-dione was prepared in dimethyl sulfoxide at a concentration of 40 mM, which was further diluted during the experiment to a final concentration of 10 μM . The holding current (I_{holding}) was measured in 50-ms epochs of traces lacking postsynaptic currents (PSCs) in control artificial cerebrospinal fluid, and in the presence of drugs and additional GABA_A blockers ($n = 40$ epochs separated by ~800 ms in each case). I_{tonic} was defined as the difference between I_{holding} before and after application of the GABA_A receptor blocker bicuculline (20 μM) (Pandit et al., 2013).

2.4. hGABA_A-receptor positive allosteric modulator IonFlux assay

Relative activities on human GABA_A receptor subtypes were studied *in vitro* using an IonFlux high-throughput (HT) electrophysiological assay. The six cell lines used were Eurofins PreciSION™ Recombinant human embryonic kidney cell lines expressing the human GABA_A α , β , and γ ion channel subunits ($\alpha_1\beta_2\gamma_2$, $\alpha_2\beta_3\gamma_2$, $\alpha_3\beta_3\gamma_2$, $\alpha_4\beta_3\gamma_2$, $\alpha_5\beta_3\gamma_2$, and $\alpha_6\beta_3\gamma_2$).

The EC₂₀ (3 μM GABA, $\alpha_1\beta_2\gamma_2$; 1 μM GABA, $\alpha_2\beta_3\gamma_2$; 2 μM GABA, $\alpha_3\beta_3\gamma_2$; 0.3 μM GABA, $\alpha_4\beta_3\gamma_2$; 0.2 μM GABA, $\alpha_5\beta_3\gamma_2$; 0.3 μM GABA, $\alpha_6\beta_3\gamma_2$) concentration of GABA was used to measure positive allosteric modulator potentiation.

Peak inward currents in response to the GABA additions in the presence of a single concentration of compound were measured. The external solution for recording hGABA_A ion channel currents on the IonFlux HT contained (in mM): 137 NaCl, 4 KCl, 1 MgCl_2 , 1.8 CaCl_2 , 10 HEPES, and 10 glucose, with a pH adjusted to 7.35 with NaOH. The internal solution included (in mM): 70 KF, 60 KCl, 15 NaCl, 5 HEPES, 5 EGTA, and 4 MgATP, with a pH adjusted to 7.3 with KOH. All compound data were normalized to the baseline peak current induced by the addition of EC₂₀ GABA for 2 s:

$$\text{Normalize Peak Current} = (I_{\text{Compound + GABA}} / I_{\text{GABA}})$$

Where $I^{\text{Compound} + \text{GABA}}$ is the peak current induced by addition of test compound + EC₂₀ GABA after 30 s incubation of test compound, I^{GABA} is the baseline peak current induced by addition of EC₂₀ GABA. All data were analyzed using Graph Pad Prism software.

2.5. Radioligand binding displacement assay

Radioligand binding displacement assays were conducted to assess the binding of cenobamate on GABA_A receptor sites. Adult Wistar rats were decapitated, the brains were immediately removed and placed in ice-cold buffer. The hippocampus, or whole brain minus cerebellum, was dissected for the binding assays. Tissues were homogenized and centrifuged, and the resulting pellets were resuspended in the corresponding buffer, as described below for each binding assay.

2.5.1. [³H]Flunitrazepam

Whole brain membranes minus cerebellum were resuspended in 50 mM phosphate buffer, pH 7.4, and incubated in duplicate with 1 nM of [³H]flunitrazepam radioligand and 10 μM of diazepam (for quantification of nonspecific binding) for 1 h at 25 °C. Under these conditions, the apparent equilibrium dissociation constant (K_d) was 4.40 nM and the maximum density of the binding sites (B_{max}) was 1.20 pmol/mg of protein. Nonspecific binding represented 9% of the total binding. The binding assay for [³H]flunitrazepam was adapted from Damm et al. (1978) and Speth et al. (1979).

2.5.2. [³H]Flumazenil

Membranes from the hippocampus were resuspended in a buffer containing 50 mM Tris-citrate, pH 7.4, and incubated in duplicate with 1 nM of [³H]flumazenil radioligand and 10 μM of diazepam (for quantification of nonspecific binding) for 2 h at 4 °C. Under these conditions, the apparent K_d was 1.20 nM and the B_{max} was 1.20 pmol/mg of protein. Nonspecific binding represented 4% of the total binding. The binding assay for [³H]flumazenil was adapted from McLeod et al. (2002) (McLeod et al., 2002).

For all assays, binding was measured in the presence of increasing concentrations of cenobamate (0.3–1000 μM). A significant response was defined as ≥50% inhibition of the radioligand binding.

2.6. Statistical analysis

Numerical values are provided as the mean ± standard error of the mean (S.E.M.). Significant differences in the mean values were tested using Student's paired two-tailed t-test or ANOVA. Values of $P < 0.05$ were considered significant.

The continuous curves for concentration response relationship of GABA or cenobamate were fitted using a least-squares fit to the following equations using a commercial program (Origin version 8.1; OriginLab and Kaleida Graph 4.5, Synergy Software);

$$I = I_{\text{max}} \times C^n / (C^n + \text{EC}_{50}^n),$$

where I is the amplitude of GABA-induced currents and C is the corresponding GABA concentration. EC₅₀ and n denote the half-effective concentration and the Hill coefficient, respectively.

2.7. Drugs

Cenobamate (YKP3089) was supplied by SK Biopharmaceuticals Co., Ltd, and other reagents used in these studies were purchased from Sigma (St. Louis, MO, USA) and/or Tocris Bioscience (Minneapolis, MN). When stock solutions of any drugs were prepared in dimethyl sulfoxide, the concentration of dimethyl sulfoxide in the final drug concentration was less than 0.33%, and this dimethyl sulfoxide concentration had no inhibitory effect on GABA_A receptor-mediated membrane currents in hippocampal CA3 neurons.

Table 1

Positive allosteric modulation of hGABA_A receptor subtypes by cenobamate.

hGABA _A subtype	EC ₅₀	% Enhancement of I_{GABA} at 37/111 μM
α ₁ β ₂ γ ₂	192 μM	31/57
α ₂ β ₃ γ ₂	119 μM	20/39
α ₃ β ₃ γ ₂	194 μM	20/45
α ₄ β ₃ γ ₂	42 μM	56/96
α ₅ β ₃ γ ₂	89 μM	11/46
α ₆ β ₃ γ ₂	58 μM	25/84

EC₅₀, half-maximal concentration; hGABA_A, human GABA_A subtypes; I_{GABA} , GABA-mediated current.

3. Results

Electrophysiological recordings were obtained from acutely isolated single CA3 neurons, CA1 pyramidal neurons of hippocampal slices, and human embryonic kidney cells expressing heterologous hGABA_A receptors.

3.1. Cenobamate positively modulates recombinant GABA_A-receptor activity

To determine whether cenobamate modulates the activity of GABA_A receptors, the effects of cenobamate on hGABA_A receptors expressed in human embryonic kidney cells were investigated. The hGABA_A receptors tested and modulated by cenobamate were composed of a different α subunit (α₁, α₂, α₃, α₄, α₅, or α₆) combined with β₂ (for α₁) or β₃ (for α₂₋₆) and a γ₂ subunit. As summarized in Table 1, cenobamate showed a positive allosteric modulation of GABA-induced (EC₂₀) current mediated by hGABA_A receptors, with similar EC₅₀ values ranging from 42 to 194 μM.

3.2. Cenobamate potentiates GABA-induced currents in a concentration-dependent manner in CA3 neurons

The effects of cenobamate on I_{GABA} in acutely isolated single CA3 pyramidal neurons were investigated. Although cenobamate itself failed to generate ionic current in the present recording conditions (data not shown), it enhanced the whole-cell currents of CA3 neurons induced by 1 μM GABA in a concentration-dependent manner (Fig. 1Aa and 1B). I_{GABA} enhancement was statistically significant at cenobamate concentrations ≥30 μM (peak I_{GABA} increased to $110 \pm 2.4\%$ of control, $n = 7$, $P < 0.05$, Student's paired two-tailed t-test). The calculated EC₅₀ value of cenobamate for I_{GABA} potentiation was 164.1 μM (Fig. 1B). Diazepam also potentiated the peak I_{GABA} in a concentration-dependent manner with an EC₅₀ value of 14.7 nM (Fig. 1Ab and 1B).

To examine the possible interaction of cenobamate with the benzodiazepine binding site, the positive modulation of cenobamate on I_{GABA} in the presence of flumazenil, a selective benzodiazepine receptor antagonist (Fig. 2Aa and 2Ba) was evaluated. Potentiation of I_{GABA} by cenobamate (100 μM) was not affected by the presence of 1 μM flumazenil ($133.1 \pm 6.6\%$ of the control in cenobamate only, and $142.0 \pm 5.2\%$ of the control in cenobamate plus flumazenil, $n = 5$, $P = 0.14$), whereas the same concentration of the benzodiazepine antagonist significantly attenuated the I_{GABA} potentiation by diazepam (10 nM) ($140.8 \pm 7.3\%$ of the control in diazepam only, and $110.3 \pm 2.5\%$ of the control in diazepam plus flumazenil, $n = 6$, $P < 0.01$) (Fig. 2Ab and 2Bb). These results suggested that cenobamate positively modulates GABA_A receptors through non-benzodiazepine binding sites.

Previous studies have suggested that potentiation of GABA currents by benzodiazepines, which offer a wide spectrum of antiepileptic activity, only occurs at sub-saturating, but not at maximally active, GABA concentrations (Choi et al., 1981; Krampfl et al., 1998; Mozrzymas et al., 2007). To determine whether cenobamate exerts its effects

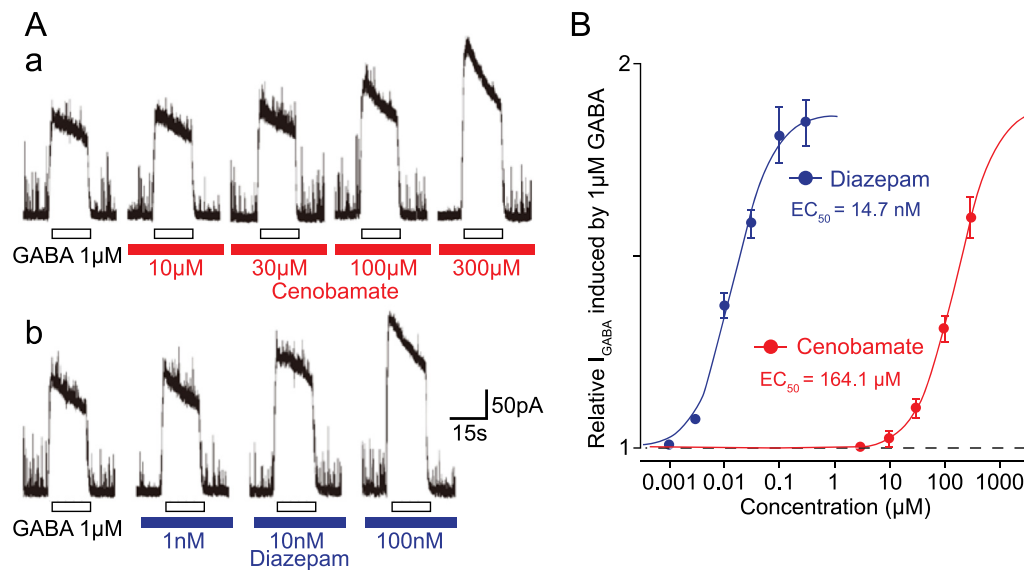


Fig. 1. Effects of cenobamate and diazepam on I_{GABA} induced by 1 μ M GABA.

A: Representative traces of I_{GABA} induced by 1 μ M GABA before, during, and after the application of various concentrations of cenobamate (a, red bars), or diazepam (b, blue bars). GABA was applied at 2–3 min intervals and cenobamate or diazepam was pre-incubated 1 min before the GABA application.

B: Concentration-potential relationship of cenobamate (red circles), or diazepam (blue circles) for the I_{GABA} induced by 1 μ M GABA. Each point was the mean $\pm$ S.E.M. from 6 to 10 experiments. The EC_{50} values of cenobamate and diazepam for the GABA (1 μ M)-induced currents were 164.1 μ M and 14.7 nM, respectively.

similarly to benzodiazepines, the effects of cenobamate (100 μ M) on I_{GABA} induced by various concentrations of GABA (0.3–300 μ M) were compared (Fig. 3).

The application of GABA elicited outward currents in a concentration-dependent manner ($EC_{50} = 8.1$ μ M, Fig. 3A and B). Cenobamate effectively potentiated I_{GABA} induced by low concentrations of GABA (1 μ M) ($132.1 \pm 9.4\%$ of the control, $n = 6$, $P < 0.01$), while it failed to facilitate I_{GABA} at GABA concentrations ≥ 30 μ M ($101.0 \pm 1.1\%$ of the control, $n = 5$, $P = 0.41$ at 100 μ M GABA) (Fig. 3A and B). The EC_{50} value was 6.8 μ M in the presence of 100 μ M cenobamate. Diazepam (100 nM) facilitated I_{GABA} at similar GABA concentration ranges in isolated CA3 neurons ($181.1 \pm 7.4\%$ of the control, $n = 5$, $P < 0.01$ at 1 μ M GABA) (Fig. 3B and C). The EC_{50} value was 5.8 μ M in the presence of 100 nM diazepam.

These results suggested that cenobamate positively modulates GABA-induced currents, which is most efficient when GABA_A receptors are exposed to a low concentration of GABA.

3.3. Cenobamate preferentially facilitates tonic over phasic GABA_A inhibition in CA1 neurons

The effects of cenobamate on I_{phasic} and I_{tonic} in hippocampal brain slices were determined. Electrophysiological recordings were obtained from a total of 43 CA1 pyramidal cells from 15 animals. Bath application of cenobamate (30–1000 μ M) induced a significant inward shift in $I_{holding}$ (I_{tonic}) in a concentration-dependent manner ($n = 7$ for 30 μ M, $n = 7$ for 100 μ M, $n = 10$ for 300 μ M, and $n = 6$ for 1000 μ M), that was blocked by the GABA_A-receptor antagonist bicuculline (BIC, 20 μ M) (Fig. 4A). I_{tonic} enhancement was statistically significant at cenobamate concentrations ≥ 30 μ M, and the calculated EC_{50} value of cenobamate for I_{tonic} was 36.63 μ M (Fig. 4B).

Low concentrations of cenobamate (30 and 100 μ M) caused minimal changes in the main characteristics of sIPSCs in the CA1 neurons ($n = 7$ for 30 μ M and $n = 7$ for 100 μ M), whereas high concentrations (300 and 1000 μ M) significantly increased the decay time constant of sIPSCs with minimal changes in the frequency and amplitudes of sIPSCs

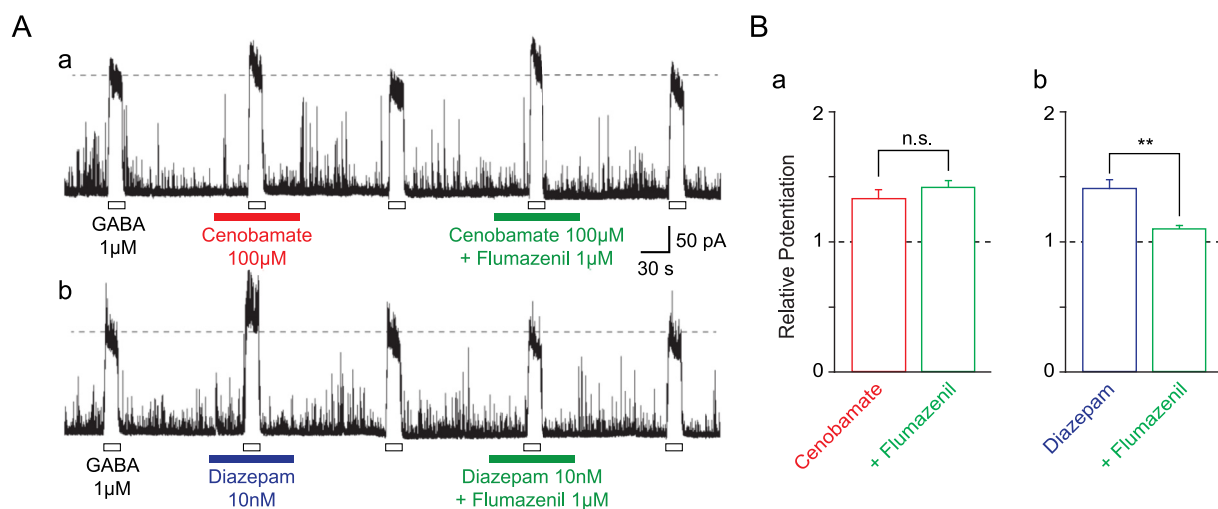


Fig. 2. Effect of flumazenil on cenobamate-mediated potentiation of I_{GABA} .

Aa: Representative traces of I_{GABA} induced by 1 μ M GABA (open bars) before, during, and after the application of 100 μ M cenobamate (red bar) or 100 μ M cenobamate plus 1 μ M flumazenil (green bar). GABA was applied at 2 min intervals and cenobamate or flumazenil was preincubated 30 s before the GABA application. **Ab:** Representative traces of I_{GABA} induced by 1 μ M GABA (open bars) before, during, and after the application of 10 nM diazepam (blue bar) or 10 nM diazepam plus 1 μ M flumazenil (green bar). GABA was applied at 2 min intervals and diazepam or flumazenil was preincubated 30 s before the GABA application. **B:** Changes in the cenobamate- (a) or diazepam-mediated (b) potentiation of GABA (1 μ M)-induced currents by flumazenil. Each column represents the mean $\pm$ S.E.M. from 5 (a, cenobamate) or 6 experiments (b, diazepam). ** $P < 0.01$, n.s.; not significant, Student's t-test.

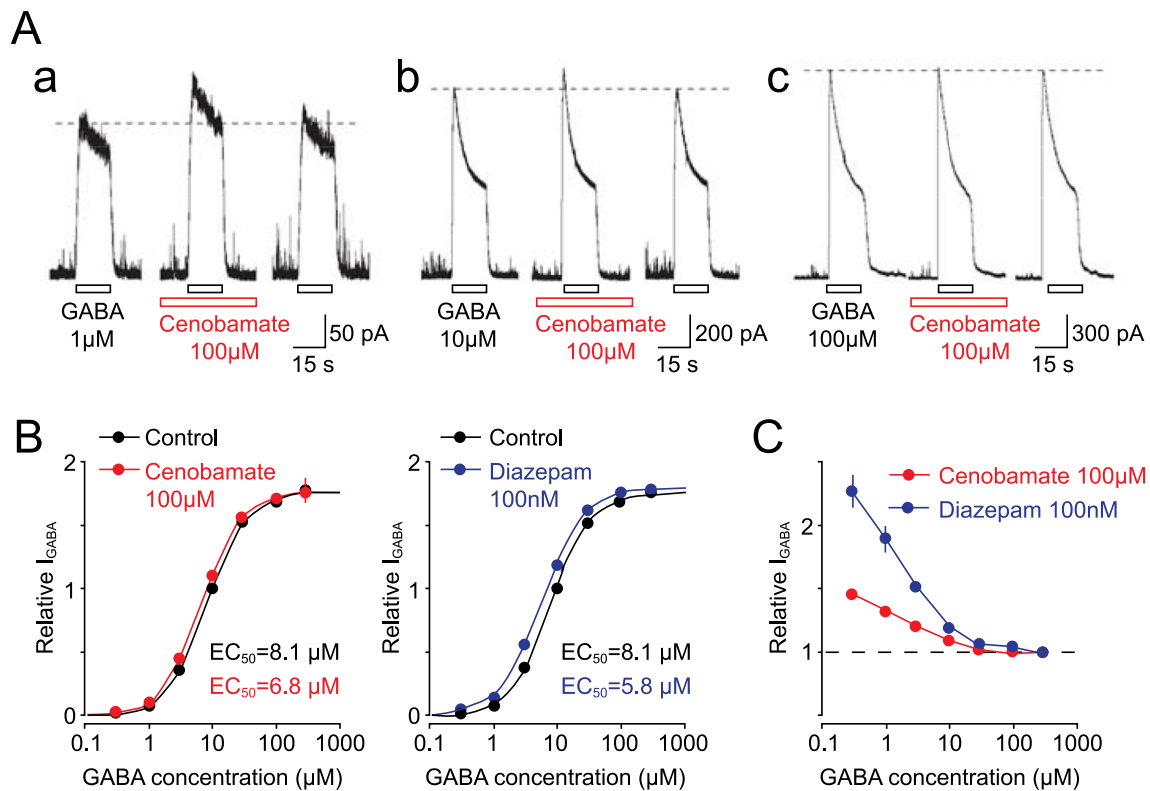


Fig. 3. Effect of cenobamate on I_{GABA} induced by various concentrations of GABA.

A: Representative traces of I_{GABA} induced by 1 μ M (a), 10 μ M (b), and 100 μ M (c) GABA (open bars) before, during, and after the application of 100 μ M cenobamate (red bars). GABA was applied at 2–4 min intervals and cenobamate was preincubated 1 min before the GABA application.

B: Concentration-inhibition relationship of cenobamate (left) and diazepam (right) for the peak I_{GABA} . All points were the mean $\pm$ S.E.M. from 6 to 10 neurons. The EC_{50} values were 8.1 μ M for GABA only, 6.8 μ M for GABA plus 100 μ M cenobamate, and 5.8 μ M for GABA plus 100 nM diazepam.

C: GABA concentration-dependent potentiation of I_{GABA} by cenobamate (red circles) or diazepam (blue circles). Each point was the mean $\pm$ S.E.M. from 6 to 10 experiments and normalized to the respective I_{GABA} .

($n = 10$ for 300 μ M and $n = 6$ for 1000 μ M) (Fig. 4C–G). The prolongation of sIPSCs was statistically significant at cenobamate concentrations ≥ 300 μ M (Fig. 4F and G).

These results suggest that while cenobamate facilitates both synaptic and extrasynaptic GABA_A-receptor activities mediating phasic and tonic GABA_A inhibition, respectively, in CA1 pyramidal neurons, cenobamate may preferentially modulate extrasynaptic GABA_A receptors over their synaptic counterparts in CA1 neurons as the low concentrations of cenobamate (30 and 100 μ M) had little effect on the amplitude and decay time of sIPSCs.

3.4. Cenobamate effects on GABA_A receptors are not blocked by benzodiazepine antagonist in CA1 pyramidal neurons

Drawing from the notion that synaptic GABA_A receptors mediating IPSCs include benzodiazepine-binding receptors (Wang, 2011) and the previous results showing that cenobamate prolonged the sIPSC decay time, it was further tested whether cenobamate may enhance GABA_A inhibition through benzodiazepine-binding receptors. The effects of flumazenil on I_{phasic} and I_{tonic} enhanced by cenobamate in CA1 pyramidal neurons were examined. Flumazenil failed to affect cenobamate-potentiated (300 μ M) I_{tonic} ($n = 4$) (Fig. 5A and B). Furthermore, flumazenil caused minimal changes in the main characteristics of sIPSCs (I_{phasic} ; $n = 4$) (Fig. 5C–G), even on the decay time of sIPSC prolonged in the presence of cenobamate ($n = 4$) (Fig. 5F and G).

In a subset of experiments, flumazenil completely reversed the diazepam-induced prolongation of sIPSC decay time to control level (control, 19.04 ± 1.56 ms, diazepam, 25.69 ± 3.09 ms, diazepam + flumazenil, 17.98 ± 1.78 ms, $n = 6$).

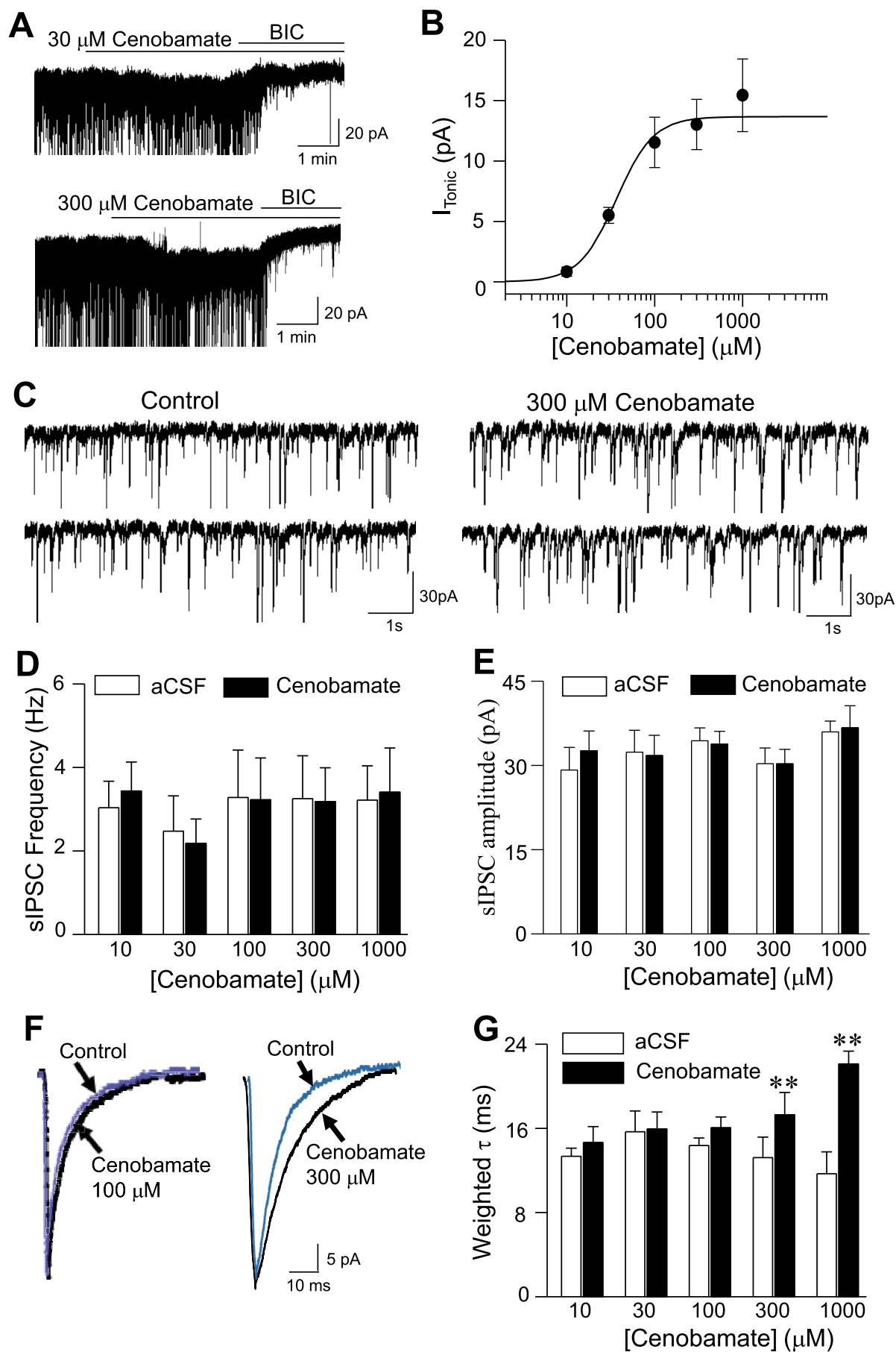
These results suggest that cenobamate does not bind to benzodiazepine sites to facilitate GABA_A inhibition in CA1 pyramidal neurons, consistent with its preferential activity on tonic inhibition. This notion was further supported by the results from radioligand displacement assays (Table 2). Cenobamate (300 μ M) did not significantly displace the binding of flunitrazepam ($\sim 4\%$ inhibition in whole brain membranes minus cerebellum) or flumazenil (0% inhibition in hippocampal membranes) to GABA_A receptors, suggesting that the positive allosteric modulation of cenobamate on GABA_A receptors does not involve benzodiazepine binding sites.

4. Discussion

While cenobamate is a newly developed antiepileptic drug undergoing late-stage clinical development, the exact antiepileptogenic mechanism of this compound has not been clearly elucidated. In the present study, the effects of cenobamate on GABA_A inhibition and its binding characteristics to GABA_A receptors were determined. Taken together, the current results showed that cenobamate efficiently enhanced tonic GABA_A inhibition via non-benzodiazepine binding sites in the principal neurons of the hippocampus, which may in part explain the underlying molecular mechanism of action for stabilizing neural circuits of the epileptic hippocampus.

4.1. Cenobamate binds to GABA_A receptors to potentiate the receptor-mediated currents

The current results using recombinant hGABA_A receptors showed that cenobamate directly binds to GABA_A receptors. The fact that



(caption on next page)

Fig. 4. Effect of cenobamate on tonic and phasic GABA_A inhibition in CA1 pyramidal cells.

A: Representative current traces showing the effect of cenobamate (30 and 300 μ M) on I_{holding} , which were blocked by 20 μ M bicuculline (BIC).

B: Mean changes in I_{holding} induced by cenobamate. Each data point represents the mean $\pm$ S.E.M. from 6 to 10 experiments. Curves were generated from the Hill equation ($I_{\text{max}} = 13.67$; $EC_{50} = 36.63$ μ M; $n_H = 2.05$).

C: Representative current traces showing the effect of cenobamate on sIPSCs from the same neuron in (A). Consecutive traces showing spontaneous GABAergic currents recorded from a holding potential of -70 mV before (upper panel) and during the application of cenobamate (lower panel).

D and E: Each column represents the mean frequency (D) and amplitude (E) of sIPSCs obtained from 6 to 10 cells.

F: Overlay of the averaged IPSCs showing the effect of cenobamate on sIPSCs from the same neuron in (A).

G: The mean decay time constant of sIPSCs before and during the application of cenobamate obtained from 6 to 10 neurons. $**P < 0.01$ compared with control (artificial cerebrospinal fluid [aCSF]).

cenobamate facilitated I_{GABA} but failed to generate any ionic currents in isolated CA3 pyramidal neurons suggests that cenobamate acts as a positive allosteric modulator of GABA_A receptors in the brain. Moreover, the minimal to no effect of cenobamate on the displacement of flunitrazepam and flumazenil binding to GABA_A receptors in the hippocampus suggests that cenobamate modulation of GABA_A receptors is through a non-benzodiazepine binding site.

Cenobamate preferentially potentiated tonic over phasic GABA_A inhibition in CA1 pyramidal neurons. Tonic GABA_A inhibition has been attributed to “spillover” of GABA from synaptic clefts following GABA-mediated IPSCs, which activates extrasynaptic GABA_A receptors in different brain regions (Glykys and Mody, 2007; Rossi and Hamann, 1998; Wu et al., 2007). Cenobamate may, therefore, potentiate I_{tonic} via increased presynaptic GABA release in the hippocampal neurons. However, the current results did not demonstrate that cenobamate potentiated I_{tonic} via altered presynaptic GABA release in CA1 neurons, as cenobamate failed to affect sIPSC frequency.

4.2. Preferential potentiation of cenobamate on tonic over phasic GABA_A-receptor currents

Our results, including the observation that benzodiazepine antagonists failed to alter cenobamate modulation of GABA_A-receptor currents (recombinant and natural), suggest that cenobamate potentiates synaptic/extrasynaptic GABA_A receptors via a non-benzodiazepine allosteric binding site. The heteropentameric composition of GABA_A receptors critically determines their pharmacology, by virtue of their subsynaptic location, as well as the biophysical properties of the receptor currents (Lee and Maguire, 2014); thus, we sought further to determine whether cenobamate displayed subtype selectivity. In general, extrasynaptic GABA_A receptors containing the δ -subunit mediate I_{tonic} (Scimemi et al., 2005; Stell et al., 2003), while their synaptic counterparts containing the γ_2 subunit mediate IPSC (Farrant and Nusser, 2005). Initial results suggested a preferential action on extrasynaptic δ -subunit-containing receptors, where cenobamate (30–100 μ M) potentiated I_{tonic} with minimal sIPSC prolongation in CA1 pyramidal neurons. However, cenobamate proved to be a more potent positive allosteric modulator of the $\alpha_4\beta_3\gamma_2$ receptor ($EC_{50} = 42$ μ M) when compared to the $\alpha_4\beta_3\delta$ receptor (300 μ M, Table 3). Improved affinity for α subunits might also confer selectivity, as GABA_A receptors are comprised of differing pairs of alpha subunits: synaptic examples contain α_1 , α_2 , or α_3 , whereas extrasynaptic examples contain α_4 , α_5 and α_6 (Farrant and Nusser, 2005; Lee and Maguire, 2014; Pavlov and Walker, 2013). Interestingly, our findings of apparent selectivity for α_4 - α_6 (lower EC_{50} values) were offset by inconsequential changes in magnitudes of enhancement (Table 1). Although these studies evaluated cenobamate against a representative set of GABA_A-receptor subtypes, they do not allow us to exclude β subunit selectivity. While a more expansive matrix of subtypes may provide additional valuable data around cenobamate interactions with GABA_A receptors, the information to date suggests that preferential action on the tonic current may not be due to GABA_A-receptor selectivity. Combined with the fact that fluctuations in tonic current (Pandit et al., 2017) and GABA_A-receptor subunit expression (Govindpani et al., 2017) have been demonstrated with increasing age, our results showing that cenobamate

consistently potentiated GABA_A-receptor-mediated current of the pyramidal neurons from 1- to 3- week-old brains also supports the conclusion that preferential action of cenobamate on the tonic current may not be due to GABA_A-receptor selectivity.

In addition to subunit selectivity, preferential facilitation of I_{tonic} at low concentrations of cenobamate may be explained by low ambient GABA concentration. For example, previous studies have shown that by increasing the affinity of the receptor for GABA through slowing the unbinding rate, benzodiazepines increase GABA_A-receptor currents differently depending on whether conductance is induced by sub-saturating (extrasynaptic conditions) or saturating (synaptic conditions) concentrations of GABA (Bianchi et al., 2009). Other studies have reported benzodiazepine-potentiated GABA responses only at sub-saturating GABA concentrations (Choi et al., 1981; Krampfl et al., 1998; Mozrzymas et al., 2007). These effects can be described by a ternary allosteric model, in which both the primary and allosteric ligands simultaneously bind to the receptor and reciprocally modulate their respective affinities. Similarly, brucine and some analogs thereof act as allosteric muscarinic receptor modulators by increasing agonist potency without affecting the maximal response (Birdsall et al., 1999; Lazareno et al., 1998). Future studies are warranted to characterize the binding and unbinding dynamics of cenobamate for its allosteric mode of action on GABA_A receptors.

4.3. Clinical significance of GABA_A-receptor modulation by cenobamate

Cenobamate is a novel neurotherapeutic compound showing broad-spectrum anticonvulsant effects in various animal models. In the present study, cenobamate showed a positive allosteric modulation for all tested recombinant hGABA_A-receptor subtypes with EC_{50} values ranging from 42 to 194 μ M, which are comparable with the steady-state concentrations of cenobamate in human plasma at clinically effective doses of 100–400 mg/day (Vernillet and Kamin). In addition, cenobamate significantly potentiated the GABA-induced current in rat hippocampal neurons at concentrations ≥ 30 μ M. Furthermore, cenobamate effectively potentiated tonic and phasic GABA_A inhibition at concentrations ≥ 30 μ M and ≥ 300 μ M, respectively, in hippocampal principal neurons. Overall, the current results demonstrate that cenobamate potentiates GABA_A-receptor inhibition in the brain, which could be a mechanism of action of an antiepileptic drug.

Cenobamate is well-tolerated and shown to be effective in patients with photosensitive epilepsy (Kasteleijn-Nolst Trenité et al., 2019). Nakamura et al. found that cenobamate modulates voltage-gated Na^+ channels that cause repetitive and/or burst generation of action potentials, predominantly via inhibition of the persistent Na^+ current (Nakamura et al., 2019). Our current study indicates that, in addition, cenobamate may enhance neuronal inhibition through modulation of GABA_A channels. A recent phase 2 clinical study reported that adjunctive cenobamate significantly improved seizure control in a dose-dependent manner in patients whose focal epilepsy was not adequately controlled with 1–3 antiepileptic drugs, an antiepileptic effect sustained for at least 12 weeks (Krauss et al., 2020). Although benzodiazepines offer a wide spectrum of antiepileptic activity against diverse types of epileptic seizures, their use in the treatment of epilepsy is limited because of tolerance resulting in desensitization and downregulation of

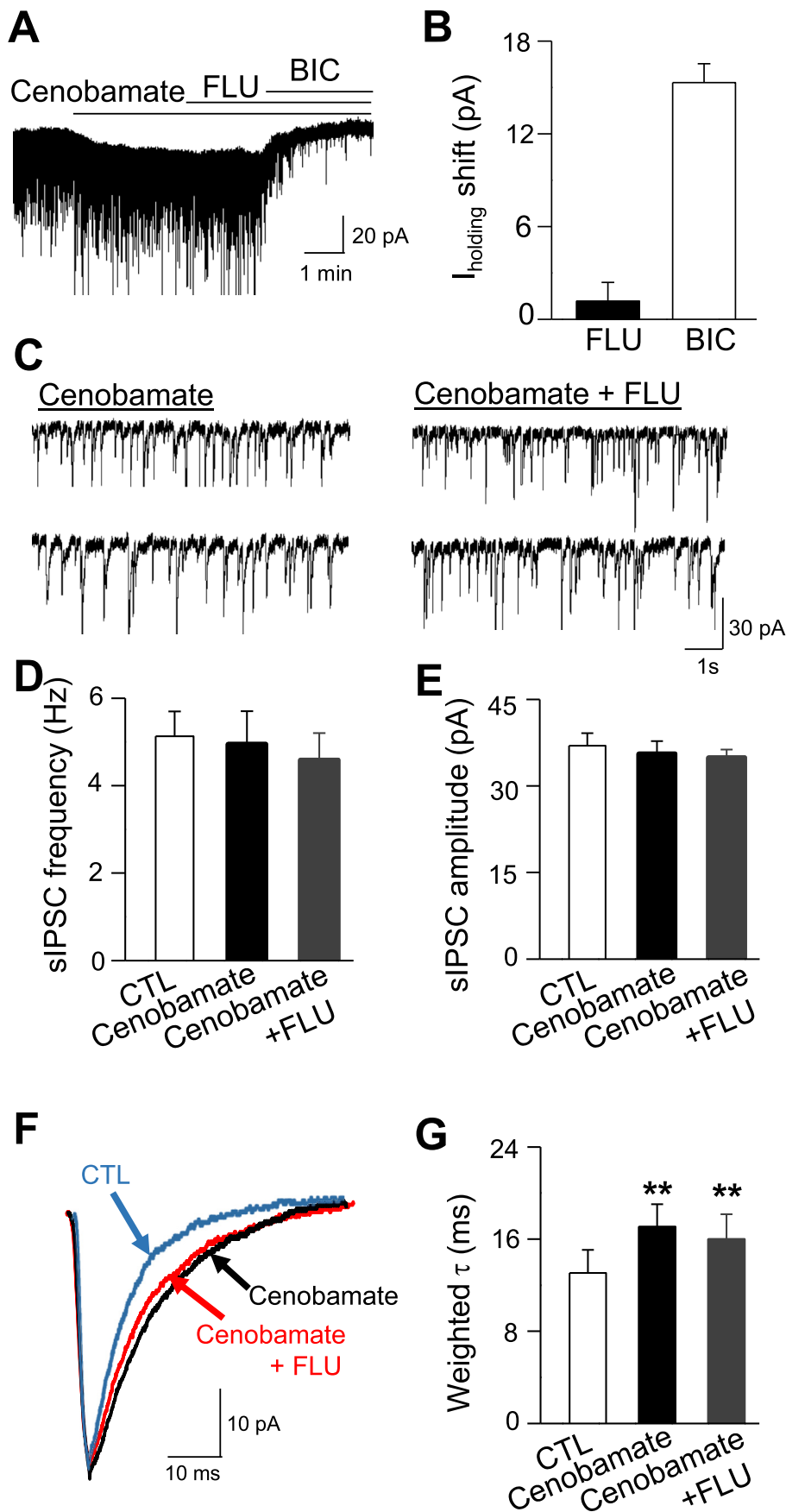


Fig. 5. Effects of flumazenil on cenobamate-potentiated tonic and phasic GABA_A inhibition.

A: Representative current traces showing the effect of cenobamate (1000 μ M) and cenobamate + flumazenil (FLU) (1 μ M) on I_{holding} .

B: Mean changes in I_{holding} induced by flumazenil and bicuculline (BIC) in the presence of cenobamate. Each column represents the mean $\pm$ S.E.M. from 7 experiments.

C: Representative current traces showing the effect of flumazenil on sIPSCs in the presence of cenobamate from the same neuron in (A). Consecutive traces showing spontaneous GABAergic currents recorded from a holding potential of -70 mV in the presence of cenobamate (left panel) and cenobamate + flumazenil (right panel).

D and E: Each column represents the mean frequency (D) and amplitude (E) of sIPSCs from control, cenobamate (300 μ M) and cenobamate + flumazenil obtained from 7 neurons.

F: Overlay of the averaged IPSCs showing the effect of flumazenil on sIPSCs prolonged by cenobamate from the same neuron in (A).

G: The mean decay time values of sIPSCs from control, cenobamate (1000 μ M) and cenobamate + flumazenil obtained from 7 neurons. $**P < 0.01$ compared with control.

ms, millisecond.

Table 2
Binding of cenobamate to GABA_A receptor: radioligand displacement assay.

Radioligand	% Inhibition of radioligand binding at 300 μ M cenobamate
Flunitrazepam	– 4
Flumazenil	0

Table 3
Positive allosteric modulation of $\alpha_4\beta_3\delta$ receptor subtype by cenobamate.

hGABA _A subtype	EC ₅₀	E _{max}	% Enhancement of I _{GABA} at 100/300 μ M
$\alpha_4\beta_3\delta$	303.02 μ M	171.17%	26.64/83.25

EC₅₀, half-maximal concentration; E_{max}, maximal concentration; hGABA_A, human GABA_A subtypes; I_{GABA}, GABA-mediated current.

GABA receptors (Miller et al., 1988). The present results indicate that cenobamate facilitates GABA_A-receptor currents via a non-benzodiazepine site on the receptors, suggesting the possibility that cenobamate may not cause tolerance development (Krauss et al., 2020).

Prior presentation/publication

Data from this analysis were previously presented in posters at the American Epilepsy Society Annual Meeting, November 30–December 4, 2018, New Orleans, LA (Sharma R et al. Poster 3.306) and at the American Academy of Neurology Annual Meeting, May 4–10, 2019, Philadelphia, PA (Sharma R et al. Poster 5–033).

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CRediT authorship contribution statement

Ramesh Sharma: Conceptualization, Data curation, Formal analysis, Validation. **Michiko Nakamura:** Data curation, Writing - original draft, Writing - review & editing. **Chiranjivi Neupane:** Conceptualization, Data curation, Formal analysis, Methodology, Writing - original draft, Writing - review & editing. **Byeong Hwa Jeon:** Conceptualization, Methodology, Resources, Validation, Writing - original draft, Writing - review & editing. **Hyewon Shin:** Conceptualization, Funding acquisition, Methodology, Project administration, Writing - original draft, Writing - review & editing. **Susan M. Melnick:** Conceptualization, Project administration, Funding acquisition, Supervision, Writing - original draft, Writing - review & editing. **Kelli J. Glenn:** Writing - original draft, Writing - review & editing. **Il-Sung Jang:** Data curation, Writing - original draft, Writing - review & editing. **Jin Bong Park:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing - original draft, Writing - review & editing.

Declaration of competing interest

RS: Declarations of interest: none.
MN: Declarations of interest: none.
CN: Declarations of interest: none.
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